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S.Y.B.Sc (Computer Science) NEP - 2023
Lab Course SEC-251-CS-P
Computer Networks

Solved Theory questions

Slip No. 1

Q1) A Write the answer of the following question (ANY 5)

[10 marks]

a) What is the purpose of a modem in a network? Can we connect directly to the internet without it?

Ans A modem is a device that allows computers, smartphones, tablets, and other devices to connect to the internet. It does this by converting digital signals from these devices into analog signals that can be transmitted over telephone lines or cable networks. Modems also do the opposite by converting incoming analog signals into digital signals so that they can be processed by your device.

A modem is an essential device in the modern world. It's an electronic device that enables computers to connect to the internet and communicate with other computers across the globe. Without a modem, you wouldn't be able to access the internet or send files between devices.

b) What is a MAC address and how is it different from an IP address?

Ans A MAC (Media Access Control) address is a unique, permanent 48-bit hardware identifier assigned to a network interface controller (NIC) by manufacturers for local network communication.

Types of MAC Address

Unicast MAC Address: Unicast MAC Address is an example of a transfer of data on the same network from a single system to another single system.

- **Multicast MAC Address:** Multicast MAC Address, as the name suggests is when data is transferred from multiple devices to multiple devices on the same network.
- **Broadcast MAC Address:** In Broadcast MAC Address, all devices within a network share data to their host systems.

The main difference between MAC and IP address is that MAC Address is used to ensure the physical address of the computer. It uniquely identifies the devices on a

network. While IP addresses are used to uniquely identifies the connection of the network with that device takes part in a network.

c) What is a loopback IP address? What is its range?

Ans Loopback refers to a communication method where the sender and receiver are the same device, allowing it to send data to itself. This concept is commonly used in networking and software development for testing and troubleshooting purposes.

A **loopback address** is a distinct reserved IP address range that starts from 127.0.0.0 and ends at 127.255.255.255.

d) Write the command to assign a static IP in Linux.

Ans **Assign IP:** sudo ip addr add [IP_ADDRESS]/[MASK] dev [INTERFACE]

Example: sudo ip addr add 192.168.1.100/24 dev eth0

The command sudo ip addr add [IP_ADDRESS]/[MASK] dev [INTERFACE] is used in Linux to **manually assign a static IP address** to a specific network interface

e) Which online tool used to check password strength?

Ans Popular and secure online tools to check password strength include

1. NordPass Password Strength Checker
2. Kaspersky's Password Check.
3. Bitwarden's Password Strength Tool

These tools instantly analyze length and complexity to estimate how long a password would take to crack.

f) How to Use Nmap to Scan for Open Ports.

Ans nmap –sP [IP Addr]

Eg. nmap –sP 192.168.5.25

Slip No. 2

Q1) A Write the answer of the following question (ANY 5)

[10 marks]

a) What is the functional difference between a switch and a router?

Ans A switch is used to connect multiple devices to create/form a network while router is used to connect multiple switch and their network to form a even lager network

b) How many hosts are possible in a /26 subnet?

Ans A full NetMask has 32 bits. The bits that are set to "1" show which part of the IP Address are network address and which part is left for host addresses.

A 26 mask means 26 of the first 32 bits are set to "1". That leaves 32–26=6 bits that are set to "0"

$2^{32-26} = 2^6 = 64$

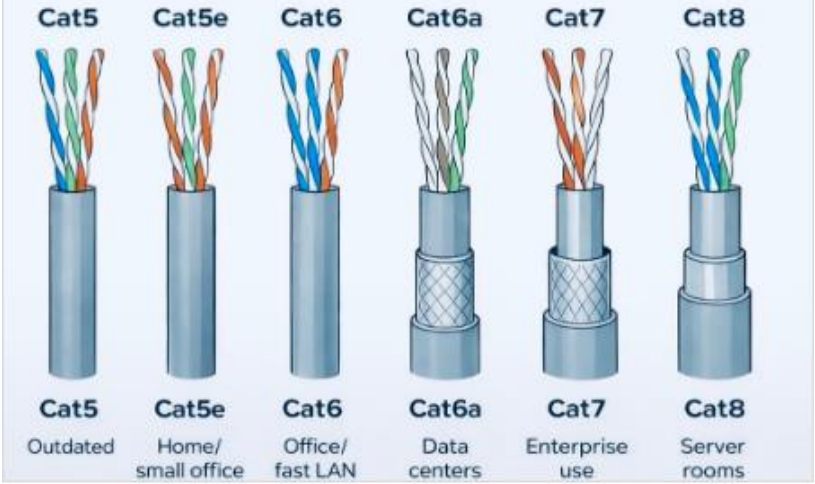
64-2=62 usable hosts (subtracting network and broadcast addresses)

The network address identifies a specific subnet (first address, host bits = 0) and cannot be assigned to hosts. The broadcast address (last address, host bits = 1) communicates with all devices in that subnet. Both are reserved and cannot be assigned to individual devices.

c) What are the differences between Cat5e, Cat6, and Cat7 Ethernet cables?

Ans

	Cat5	Cat5e	Cat6	Cat6a	Cat8
Speed	Up to 100 Mbps	Up to 1 Gbps	Up to 10 Gbps	Up to 10 Gbps	Up to 40 Gbps
Bandwidth	100 MHz	100 MHz	250 MHz	250 MHz (600–1000 MHz)	2000 MHz
Max Distance	100 m	100 m	100 m (10 Gbps)	100 m	30 m
Shielding	No	Optional	Optional	Yes	Yes
Common Use	Outdated	Home/ small office	Yes	Yes	Yes



d) Write the steps to identify a phishing email.

Ans

1. Check the Email Address

Phishers often hide their real email addresses to look like they're from reputable sources. Pay attention to slight variations in the sender's email address.

- **Legitimate emails**: From well-known domains like @bank.com or @amazon.com.
- **Phishing attempt**: Look for unusual domain names like @service-support.com or @amazonsupport.net. The domain name might be similar to a legitimate domain but with an extra character or misspelled.

2. Suspicious or Generic Greeting

Phishing emails often use a generic greeting such as "Dear Customer" or "Dear User" because the attacker does not know your actual Name or Gender.

- **Legitimate emails**: Most legitimate companies address you by your name (e.g., "Dear John Doe").
- **Phishing attempt**: Generic greetings or misspellings in the salutation are red flags.

Example:

- ***Phishing****: "Dear Customer, your account has been compromised."*
- ***Legitimate****: "Dear John Doe, we've noticed unusual activity on your account."*

3. Sense of Urgency or Threat

Phishing emails often create a sense of urgency or fear to prompt quick action. For example, they might threaten that your account will be locked unless you take action immediately, or you risk losing a reward.

- **Legitimate emails**: Legitimate companies usually don't pressure you to act right away.
- **Phishing attempt**: Emails that demand immediate action to avoid penalties or losses are highly suspicious.

Example:

- ***Phishing****: "Your account has been compromised. Click here immediately to avoid your account being locked."*
- ***Legitimate****: "We've detected some unusual activity. Please review your account when convenient."*

4. Suspicious Links or Attachments

Phishing emails often include suspicious links or attachments designed to install malware on your device or steal your information. Hover over any link to check its destination URL.

- **Legitimate emails**: Links in legitimate emails lead to official websites (e.g., https://www.amazon.com/).
- **Phishing attempt**: Hovering over a link might show a different URL than expected, such as http://www.maliciousite.com.

Example:

- ***Phishing****: A link that looks like www.yourbank-secure.com, but actually redirects to www.maliciousbank.com.*
- ***Legitimate****: A direct link to the official website (e.g., www.yourbank.com).*

5. Grammar and Spelling Mistakes

Many phishing emails contain noticeable spelling or grammatical errors. These can often be overlooked, but paying attention to these small mistakes can help you **recognize a phishing email**.

- **Legitimate emails**: Professional emails from reputable companies are proofread and free of errors.
- **Phishing attempt**: Phishing emails may have awkward phrasing, incorrect spelling, or missing punctuation.

Example:

- ***Phishing****: "Your accunt is suspecious, click here to secure your profile."*
- ***Legitimate****: "Your account has been flagged for suspicious activity. Please review your account details."*

e) What are the two analysis methods used for analyzing malware?

Ans **Static Analysis**

It is the method of examine malware without executing it. This is typically done by analyzing the code of a binary file to understand its functionality and identify any malicious activity. Static analysis is use to identify potential security threats in a sample without the risk of infecting the analysis environment. It is often used as a first step in malware analysis, to gather initial information and identify potentially malicious code before dynamic analysis is performed.

Dynamic Analysis

In this method, the malware sample is executed in controlled environment, such as sandbox, and its behavior is observed interactively. It involves monitoring the file system, registry, and network activity of the malware as it runs, and observing any malicious behavior, such as data exfiltration or unauthorized connections to remote servers.

f) What is the use of the ping command in networking?

Ans It is used to test the availability of the device in the network

Slip No. 3

Q1) A Write the answer of the following question (ANY 5)

[10 marks]

a) What does the command tracert or traceroute do?

Ans Traceroute is a network diagnostic tool used to track the path that data packets take from one computer to another across a network (like the internet). It helps in determining the route and measuring the transit delays of packets over the internet. It's especially useful for identifying network issues such as slow response times, network congestion, or routing problems.

In short:
It allows network administrators, IT professionals, and even regular users to diagnose problems with their internet connection or the websites they are trying to reach.

Eg. tracert www.google.com

Traceroute Output Example

Here is what a typical Traceroute output might look like:

traceroute to google.com (142.250.190.78), 30 hops max, 60 byte packets
1 192.168.1.1 (192.168.1.1) 1.111 ms 0.839 ms 0.712 ms
2 10.50.1.1 (10.50.1.1) 5.458 ms 5.283 ms 5.089 ms
3 72.14.205.69 (72.14.205.69) 20.274 ms 21.180 ms 20.411 ms

- **1, 2, 3, etc.** are the hops (routers).
- The **IP addresses** of the routers are displayed.
- The **time** (in milliseconds) for each hop is shown. If the same hop has multiple times, it means the packet was sent multiple times, and these represent the round-trip time.
- If a hop doesn't respond, it is marked with an asterisk *, indicating a timeout or that the router is not responding to ICMP requests.

b) Differentiate between ip addr and ifconfig commands in Linux.

Ans **ip addr**

- Displays and manages:
  - IP addresses (IPv4 & IPv6)
  - Network interfaces

ip addr → Full IPv6 support

ifconfig

- Displays and configures:
  - IP address
  - Netmask
  - Broadcast address

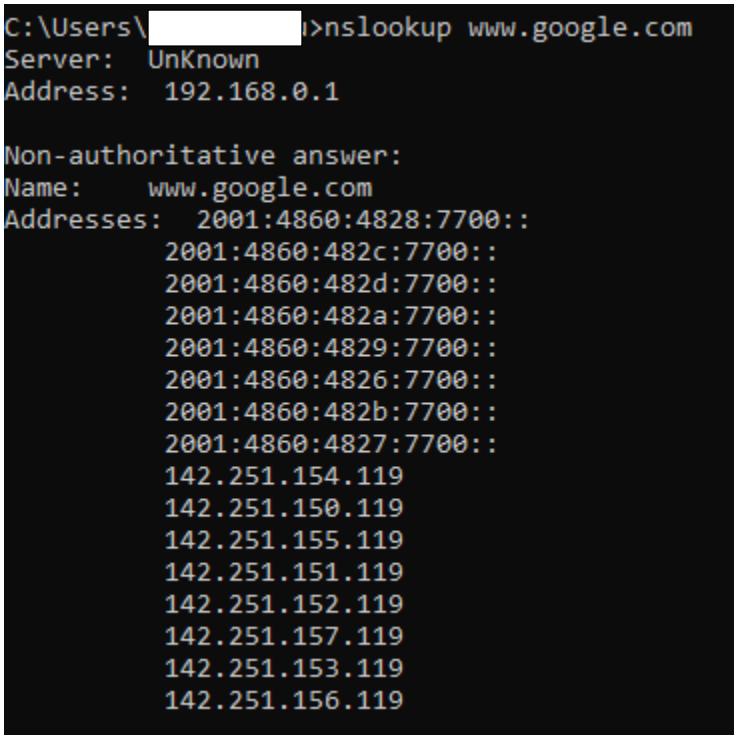
ifconfig → Limited IPv6 support

c) What does nslookup do?

Ans Nslookup, which stands for "Name Server Lookup," is a command-line tool used to query Domain Name System (DNS) servers. It helps users obtain information about domain names, IP (Internet Protocol) addresses, and DNS records.

Eg. nslookup www.google.com

Output window



```
C:\Users\[redacted]>nslookup www.google.com
Server:  UnKnown
Address:  192.168.0.1

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:4828:7700::
          2001:4860:482c:7700::
          2001:4860:482d:7700::
          2001:4860:482a:7700::
          2001:4860:4829:7700::
          2001:4860:4826:7700::
          2001:4860:482b:7700::
          2001:4860:4827:7700::
          142.251.154.119
          142.251.150.119
          142.251.155.119
          142.251.151.119
          142.251.152.119
          142.251.157.119
          142.251.153.119
          142.251.156.119
```

d) What is the function of a switch in a computer network?

Ans It is a device that enables the devices like computers, printers and servers to communicate within the network.

e) What is network topology? What are its applications?

Ans It is an arrangement or layout of the devices connected in the network. It defines how computers, switches, routers, and other peripherals are connected and how data flows between them, directly impacting performance, reliability, scalability, and security.

Types of topologies:

1. Star
2. Mesh
3. Tree

4. Bus
5. Ring
6. Hybrid

Applications of Network Topology:

i. **IT Infrastructure Design and planning**
ii. **Network Troubleshooting and maintainence**
iii. **Security Management**
iv. **Scaling and Upgrading**

f) List two advantages and two disadvantages of dynamic IP configuration.

Ans **Advantages:**

i. **Automatic Configuration:** Simplifies network administration by automatically assigning IP addresses, removing the need for manual configuration on each device.
ii. **Efficient IP Management:** Enables IP address reuse, allowing a limited pool of addresses to serve a larger number of devices that are not all connected simultaneously.

Disadvantages:

i. **Unsuitable for Hosting:** Because the IP address changes periodically, it is not ideal for hosting servers, websites, or VPNs that require a consistent address.
ii. **Potential for Connectivity Issues:** If the DHCP server fails or fails to renew a lease, devices may lose network connectivity.

Slip No. 4

Q1) A Write the answer of the following question (ANY 5)

[10 marks]

a) What is Cyber Attack? Write the types of Cyberattacks.

Ans A cyberattack is a deliberate attempt by an individual or group to breach, damage, or disrupt computer systems, networks, or digital devices, often for malicious purposes such as data theft, espionage, financial fraud, or system sabotage.

- **Phishing**
- **Ransomware**
- **Malware**
- **Denial-of-Service (DoS/DDoS)**
- **SQL Injection (SQLi**
- **Man-in-the-Middle (MitM) Attack**
- **Brute-Force Attack**
- **Zero-Day Exploit**.
- **Social Engineering**

b) How can you reset or release an IP address obtained from DHCP?

Ans In Windows, use `ipconfig /release` followed by `ipconfig /renew` in CMD

Releasing a DHCP IP address is the process of a client device notifying the DHCP server that it no longer needs its current IP address, allowing it to be returned to the available pool for other devices. It is often followed by a "renew" command to obtain a fresh IP, commonly used for resolving network connectivity issues.

c) Explain supernetting with an example combining IPv4 Class C networks.

Ans

Combining Four Class C Networks

Imagine a company with four Class C networks:

a. 192.168.0.0/24
b. 192.168.1.0/24
c. 192.168.2.0/24
d. 192.168.3.0/24

1. Analyze the Binary

To combine them, analyze the third octet of the network addresses in binary:

0 = 00000000
1 = 00000001
2 = 00000010
3 = 00000011

The first 6 bits (000000) are common across all four networks, meaning we can combine them by shifting the subnet boundary 2 bits to the left, from /24 to /22

2. Determine the Supernet Address

New subnet mask: /22 (22 ones and 10 zeros), which is 255.255.252.0

The supernet address: 192.168.0.0/22 (covers 192.168.0.0 through 192.168.3.255)

d) Explain the use of copy running-config startup-config.

Ans Cisco devices store commands in two configuration files:

startup configuration
running configuration

Immediately after you type a command in the global configuration mode, it will be stored in the running configuration. A running configuration resides in a device's RAM, so if a device loses power, all configured commands will be lost.

To avoid this scenario, you need to copy your current configuration into the startup configuration. A startup configuration is stored in the nonvolatile memory of a device, which means that all configuration changes are saved even if the device loses power.

To copy your running configuration into the startup configuration you need to type the command *copy running-configuration startup-configuration*.

e) What is a loopback IP address? What is its range?

Ans Refer Slip No. 1

Slip No. 5

Q1) A Write the answer of the following question (ANY 5)

[10 marks]

a) What does the ip link set eth0 up command do?

Ans ip link set eth0 up is used in Linux to **enable (activate) a network interface**.

b) How to detect hidden Wi-Fi networks

Ans By default, all WiFi networks broadcast the so-called Service Set Identifier (SSID) every 100 milliseconds to let other devices know about their presence. Some routers don't broadcast their SSID because the hidden network WiFi option has been turned on in the settings. This option stops the broadcasting of the network's SSID, which effectively causes it to disappear and become invisible to all devices attempting to detect Wi-Fi signals. The SSID is the unique name assigned to a wireless network. It is used to identify the network and differentiate it from other networks in the area. To obtain the SSID of a hidden network, you will need to have the exact name provided by the network administrator. Alternatively, you can use a network analyzer to capture the name as its being broadcast.

Encryption is a security measure used to protect wireless networks from unauthorized access. There are several types of encryption available, including WEP, WPA **(Wi-Fi Protected Access)**, WPA2 **(WPA version 2)**, and the latest standard, WPA3 **(WPA version 3)**. WEP **(Wired Equivalent Privacy)** is an older encryption standard that is no longer considered secure, while WPA2 is still widely used and recommended. WPA3 is the newest and most secure standard, offering enhanced protection against password guessing attacks and other security threats.

c) Write the command sequence to configure a router interface with IP.

Ans

```
enable
configure terminal
interface <interface-name>
ip address <ip-address> <subnet-mask>
no shutdown
exit
```

d) What is the purpose of assigning a hostname to a switch?

Ans Assigning a hostname to a switch provides a unique, human-readable identifier (e.g., "Core-Switch-01") that distinguishes it from other network devices, making management and identification easier.

e) What is the ethical hacking?

Ans Ethical hacking is the practice of legally testing computer systems, networks, or applications to find security vulnerabilities before malicious attackers can exploit them.

f) What does the show running-config command display?

Ans The show running-config command is used in networking devices (like routers and switches) to display the current active configuration stored in RAM**.**

Slip No. 6

Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) List different types of network cables and their uses of Coaxial, Twisted Pair (STP & UTP),Fiber Optic

Ans

Medium	Use Case
Coaxial	TV, CCTV
UTP	LAN
STP	Data centers
Fiber Optic	Internet backbone

b) What is the role of no shutdown in router configuration?

Ans The no shutdown` command in Cisco IOS is used to enable or activate a router interface.

c) Compare the output of ip addr and ifconfig

Ans

- ip addr provides more **detailed, structured, and modern output**.
- ifconfig provides **basic information with additional packet statistics** but is now outdated.

d) What command do you type to move from Router> to Router# mode?

Ans Cisco IOS uses a hierarchical command-line interface (CLI) divided into several distinct modes. Each mode provides a specific set of commands for monitoring or configuring the device.

Command modes:

i. User EXEC Mode
ii. Privileged EXEC Mode
iii. Global Configuration Mode
When Router> command is used it in EXEC mode.
When Router# command is used it in Privileged EXEC mode

To move form from Router> to Router# mode **enable command** is used.

e) How many hexadecimal digits are needed for a 64-bit WEP key? Give any 5 examples.

Ans A 64-bit WEP key requires exactly **10 hexadecimal digits**. These digits can include the numbers 0–9 and letters a–f (or A–F).

5 Examples of 64-bit WEP Keys:

A4583DFF1A
1A648C9FE2
0A5C1B6D8E
F9E23D4A8C
0011223344

Slip No. 7

Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) Explain the difference between ping and traceroute.

Ans Ping checks if a destination is reachable and measures latency (round-trip time) using ICMP packets, while traceroute maps the entire network path, identifying each router hop between the source and destination.

b) Explain steps to identify a phishing email.

Ans Refer slip no. 2

c) What is the function of a switch in a computer network?

Ans Refer slip no. 3

d) How many hexadecimal digits are needed for a 64-bit WEP key? Give any 5 examples.

Ans Refer slip no. 6

e) What is the primary purpose of a banner motd? How can it serve both a legal and security function?

Ans The primary purpose of a banner motd (Message of the Day) is to display a temporary, informational message to all users—authorized or unauthorized—upon their initial connection to a network device (like a router or switch) before authentication. It serves both legal and security functions by warning against unauthorized access and notifying users of monitoring policies

Slip No. 8

Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) Why is it important to configure an enable (EXEC mode) password on a switch?

Ans Configuring an enable (EXEC mode) password on a switch is crucial for network security. It acts as a secondary layer of protection that restricts unauthorized access to privileged commands, preventing unauthorized users from modifying device configurations, viewing sensitive data, or disrupting network operations.

b) Explain the role of VLAN1 in a switch's initial configuration.

Ans **VLAN 1** is the **default VLAN** on most network switches (especially Cisco switches) and plays an important role when the switch is first powered on.

All switch ports are **assigned to VLAN 1 by default**.

This means: Any device connected initially can communicate within the same network (broadcast domain).

c) Define Hashing.

Ans Hashing is the process of transforming any given key or a string of characters into another value. This is usually represented by a shorter, fixed-length value or key that represents and makes it easier to find or employ the original string.

Algorithms (like SHA-256 or MD5) that convert data into a numeric value.

d) What are Cyber Security Policies.

Ans A cyber security policy provides guidance to an organization's employees on how to act to protect the company's sensitive information. Companies commonly have several security policies that cover various topics, including IT security, email security, and the use of personal devices for work

Slip No. 9

Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) List and explain all five IP address classes with examples.

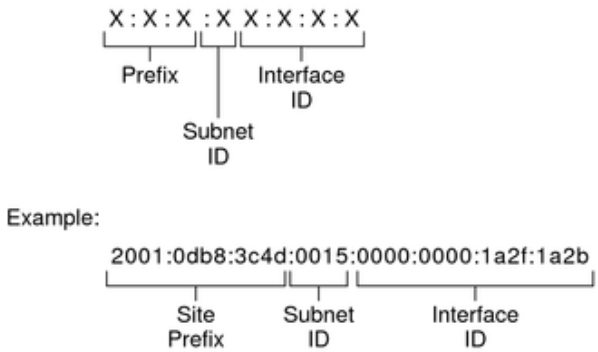
Ans

Five Different Classes of IPv4 Addresses

Class	First Octet decimal (range)	First Octet binary (range)	IP range	Subnet Mask	Hosts per Network ID	# of networks
Class A	0 – 127	0XXXXXXX	0.0.0.0-127.255.255.255	255.0.0.0	$2^{24}-2$	2^{7}
Class B	128 – 191	10XXXXXX	128.0.0.0-191.255.255.255	255.255.0.0	$2^{16}-2$	2^{14}
Class C	192 – 223	110XXXXX	192.0.0.0-223.255.255.255	255.255.255.0	$2^{8}-2$	2^{21}
Class D (Multicast)	224 – 239	1110XXXX	224.0.0.0-239.255.255.255			
Class E (Experimental)	240 – 255	1111XXXX	240.0.0.0-255.255.255.255			

b) Explain the format and example of an IPv6 address

Ans



c) Define Cyber Security.

Ans **Cybersecurity** is the practice of protecting people, systems and data from cyberattacks by using various technologies, processes and policies.

d) Explain Ethical Hacking.

Ans Ethical hacking is the use of hacking techniques by friendly parties in an attempt to uncover, understand and fix security vulnerabilities in a network or computer system.

Ethical hackers have the same skills and use the same tools and tactics as malicious hackers, but their goal is always to improve network security without harming the network or its users.

e) Differentiate between Static NAT, Dynamic NAT, and Port Address Translation (PAT)

Ans Static NAT, Dynamic NAT, and Port Address Translation (PAT) differ primarily in how they map private IP addresses to public ones.

Static NAT: Maps one private IP to one public IP consistently. Used for servers that must be reachable from the internet, such as web or email servers.

Dynamic NAT: Maps internal IPs to a pool of available public IP addresses. It is used in situations where internal users need internet access but do not need a fixed public IP.**PAT (NAT Overload):** Maps multiple private IPs to a single public IP, differentiating sessions by port numbers. It is the most common form, used in home routers to conserve public IPv4 addresses.

Slip No. 10

Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) What are the differences between Cat5e, Cat6, and Cat7 Ethernet cables?

Ans Refer Slip No. 2

b) What is the purpose of a modem in a network? Can we connect directly to the internet without it?

Ans Refer Slip No. 1

c) List different types of network cables and their uses.

Ans Refer Slip No. 2

d) Differentiate between Static NAT, Dynamic NAT, and Port Address Translation (PAT).

Ans Refer Slip No. 9

e) Write the command to assign a static IP in Linux.

Ans Refer Slip No. 1

Slip No. 11

Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) Compare and contrast the functionalities of a switch and a router.

Ans **A switch connects multiple devices to create a network**, a router connects multiple switches, and their respective networks, to form an even larger network.

b) What is the purpose of a modem in a network? Can we connect directly to the internet without it?

Ans Refer Slip No. 1

c) List different types of network cables and their uses.

Ans Network cables connect devices to networks and the internet, with main types including
Twisted Pair (UTP/STP)
Fiber Optic
Coaxial cables.
Common Ethernet categories range from Cat5e (1 Gbps) to Cat8 (40 Gbps), used for home networking, data centers, and industrial automation to ensure stable and high-speed data transmission

d) Define IPv4 and IPv6. Mention key differences between them.

Ans IPv4 uses a 32-bit address format and can accommodate more than 4 billion address spaces. With the expansion of the internet and Internet of Things (IoT) systems, IPv4 is proving to be insufficient in its addressing range. It is being phased out by IPv6, which uses a 128-bit address format and can accommodate more than $1x10^{36}$ addresses.

e) What is subnetting? How many hosts are possible in a /26 subnet?

Ans Refer Slip No. 1

f) What is the purpose of assigning a hostname to a switch?

Ans Refer Slip No. 5

Slip No. 12

Q1) A Write the answer of the following question (ANY 5) [10 marks]

a) What is the function of a switch in a computer network?

Ans Refer Slip No. 3

b) Write the command to assign a static IP in Linux.

Ans Refer Slip No. 1

c) What does the ip link set eth0 up command do?

Ans Refer Slip No. 5

d) Explain the difference between ping and traceroute.

Ans Refer Slip No. 7

e) What does the ip route show command display?

Ans The ip route show command (or simply ip route) is used to display the **IP routing table** of a network device or server.